

Chapter 5.8: A Common Framework for Memory Hierarchy

The Four Questions and the Three Cs

Contents

1 Introduction: A Unified View

Although different types of memory hierarchies (caches, TLBs, virtual memory) differ **quantitatively**, they share many **qualitative** policies and features.

All memory hierarchies can be understood by answering **four fundamental questions**:

1. **Where** can a block be placed?
2. **How** is a block found?
3. **Which** block should be replaced on a miss?
4. **What** happens on a write?

2 Question 1: Where Can a Block Be Placed?

Block Placement Options

Scheme	Number of Sets	Blocks per Set
Direct Mapped	# of blocks	1
n-way Set Associative	# of blocks / n	n
Fully Associative	1	# of blocks

2.1 Associativity Trade-offs

- **Advantage:** Increasing associativity **decreases miss rate**
 - Moving from direct-mapped to 2-way: **20–30% reduction** in miss rate
 - Diminishing returns beyond 4-way
- **Disadvantages:** Increased cost and **slower access time**

Cache Size Impact

- As cache sizes grow, the **relative improvement** from associativity increases only slightly
- The **absolute improvement** shrinks significantly (less opportunity)
- Smaller caches benefit more from associativity

3 Question 2: How Is a Block Found?

Block Location Methods

Scheme	Location Method	Comparisons
Direct Mapped	Indexing	1
Set Associative	Index set, then search	n (degree of assoc.)
Fully Associative	Full search of all entries	All blocks
Virtual Memory	Separate page table	0 (table lookup)

3.1 Why Fully Associative for Virtual Memory?

Virtual memory systems **almost always** use fully associative placement because:

1. **Misses are very expensive** (disk access!) – full associativity minimizes misses
2. **Software replacement schemes** can be more sophisticated
3. The page table provides **easy indexing** with no extra hardware search

3.2 Direct-Mapped Advantage

Direct-mapped caches have a key advantage: **faster access time and simplicity** because finding the block does not depend on a comparison – just indexing.

4 Question 3: Which Block Should Be Replaced?

Replacement Strategies

- **Direct Mapped:** No choice – only one candidate
- **Associative Caches:** Must choose from set (or entire cache)

4.1 Replacement Policies

Random Replacement

- Candidate blocks are selected **randomly**
- Simple to implement in hardware
- For 2-way caches: miss rate $\approx 1.1\times$ higher than LRU

Least Recently Used (LRU)

The block replaced is the one that has been **unused for the longest time**.

- Difficult to implement for high associativity (>4 -way)
- Often **approximated** (e.g., track LRU pair, then LRU within pairs)
- Large caches: random can sometimes outperform simple LRU approximations

4.2 Virtual Memory Replacement

In virtual memory, some form of **LRU is always approximated** because:

- Cost of a miss is **enormous** (disk access)
- Even tiny reduction in miss rate is valuable
- **Reference bits** help OS track less recently used pages

5 Question 4: What Happens on a Write?

Write Policies

Write-Through	Write to both cache and lower-level memory
Write-Back	Write only to cache ; write to lower level when block is replaced

5.1 Write-Back Advantages

- Processor can write at **cache speed** (not memory speed)
- Multiple writes to same block $\rightarrow$ **only one memory write**
- Enables **high-bandwidth transfers** (write entire block at once)

5.2 Write-Through Advantages

- **Misses are simpler** – never need to write back a dirty block first
- **Easier to implement** (though still needs write buffer)

5.3 What's Used in Practice?

- **Virtual Memory:** Always write-back (disk latency is too long)
- **Lowest-Level Caches:** Typically write-back (processor generates writes faster than memory can accept)

6 The Big Picture: Four Questions Summary

The Big Picture: Common Framework

Question 1: Where can a block be placed?	Direct mapped (one place), set associative (few places), or fully associative (any place)
Question 2: How is a block found?	Indexing (direct), limited search (set assoc.), full search (fully assoc.), or page table (VM)
Question 3: What block is replaced?	LRU or Random
Question 4: How are writes handled?	Write-through or write-back

7 The Three Cs Model

The **Three Cs Model** classifies all cache misses into three categories:

Compulsory Misses (Cold-Start Misses)

Misses caused by the **first access** to a block that has never been in the cache.
Reduce by: Increasing block size (fewer blocks to touch)

Capacity Misses

Misses that occur because the cache **cannot contain all blocks** needed during execution.
Reduce by: Increasing cache size

Conflict Misses (Collision Misses)

Misses in direct-mapped or set-associative caches when multiple blocks **compete for the same set**. These are misses that **would be eliminated** in a fully associative cache of the same size.
Reduce by: Increasing associativity

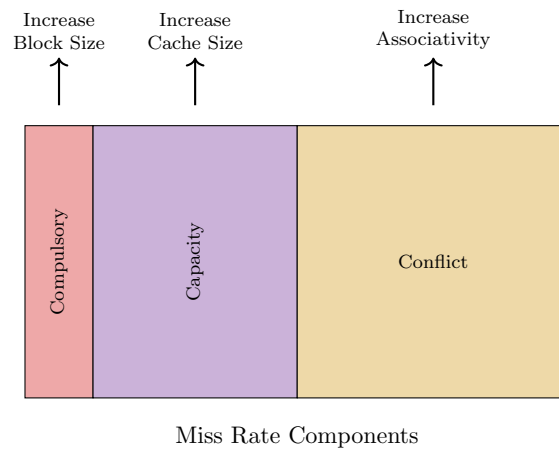


Figure 1: The Three Cs and how to reduce each type of miss.

7.1 Trade-offs: Every Improvement Has a Cost

The Challenge of Memory Hierarchy Design

Every change that potentially **improves miss rate** can also **negatively affect performance**:

Change	Benefit	Potential Cost
Increase block size	Fewer compulsory misses	Higher miss penalty
Increase cache size	Fewer capacity misses	Higher access time
Increase associativity	Fewer conflict misses	Higher access time

7.2 Key Observations from Three Cs Analysis

- **Compulsory misses** are typically very small (often <1%)
- **Conflict misses** can be significant for direct-mapped caches
- Going from 4-way to 8-way shows **very little improvement**
- **Fully associative caches have NO conflict misses** (by definition)

8 Check Yourself

Which of the following statements is generally true?

1. **There is no way to reduce compulsory misses.**
FALSE – Increasing block size reduces compulsory misses.
2. **Fully associative caches have no conflict misses.**
TRUE – By definition, conflict misses are those eliminated by a fully associative cache.
3. **In reducing misses, associativity is more important than capacity.**
Not generally true – depends on the workload and cache configuration.

9 Flashcards

9.1 From Flashcards (20)

1. **Q: What are the four fundamental questions for understanding any memory hierarchy?**
A: 1. Where can a block be placed? 2. How is a block found? 3. Which block should be replaced on a miss? 4. How are writes handled?
2. **Q: What is the term for a placement scheme where a block can be placed in exactly one location?**
A: Direct mapped.
3. **Q: In a ----- placement scheme, a block can be placed in a restricted set of places.**
A: Set-associative.
4. **Q: What placement scheme allows a block to be placed anywhere in the upper level?**
A: Fully associative.
5. **Q: What is the primary advantage of increasing the degree of associativity?**
A: It usually decreases the miss rate by reducing misses that compete for the same location.
6. **Q: What is the typical percentage reduction in miss rate from direct-mapped to 2-way set associative?**
A: Between 20% and 30%.
7. **Q: What are the two potential disadvantages of increasing associativity?**
A: Increased cost and slower access time.
8. **Q: Why does the absolute improvement from associativity shrink as cache sizes increase?**
A: Because the overall miss rate of a larger cache is already lower, leaving less opportunity for improvement.
9. **Q: How is a block found in a direct-mapped cache?**
A: By using an index, which requires only one comparison.
10. **Q: How is a block located in a set-associative cache?**
A: By first indexing the set and then searching among the elements within that set.
11. **Q: What type of search is required to find a block in a fully associative cache?**
A: A full search of all cache entries.
12. **Q: What data structure is used to find a block in a virtual memory system?**
A: A separate lookup table, specifically the page table.
13. **Q: Why are fully associative caches prohibitive except for small sizes?**
A: The cost of the comparators required for a full search becomes overwhelming for large caches.
14. **Q: What block placement scheme is almost always used for virtual memory systems?**
A: Fully associative placement.
15. **Q: What are the three motivating factors for fully associative placement in virtual memory?**
A: Misses are very expensive, it allows for sophisticated software replacement schemes, and the map can be easily indexed.
16. **Q: What is the primary advantage of a direct-mapped cache design?**
A: Faster access time and simplicity, as finding the block does not depend on a comparison.
17. **Q: In an associative cache, which replacement strategy involves selecting a candidate block at random?**
A: Random replacement.
18. **Q: What does the LRU replacement strategy dictate?**
A: The block that has been unused for the longest time should be replaced.
19. **Q: Why is true LRU costly to implement for caches with high associativity (>4-way)?**
A: Because tracking the detailed usage information for many blocks is expensive in hardware.

20. **Q: For a 2-way set-associative cache, how does random replacement compare to LRU?**
A: The miss rate for random replacement is about 1.1 times higher than for LRU.
21. **Q: What type of replacement algorithm is always approximated in virtual memory, and why?**
A: Some form of LRU is always approximated because the cost of a miss is enormous.
22. **Q: What is a write-through policy?**
A: Information is written to both the block in the cache and the block in the lower level simultaneously.
23. **Q: What is a write-back policy?**
A: Information is written only to the block in the cache, and the modified block is written to the lower level only when replaced.
24. **Q: What are the three key advantages of write-back?**
A: Processor can write at cache speed, multiple writes require only one lower-level write, and it allows for high-bandwidth transfers.
25. **Q: What are the two main advantages of write-through?**
A: Misses are simpler and cheaper, and the policy is easier to implement.
26. **Q: Why is write-back the only practical option for virtual memory systems?**
A: Because of the very long latency of a write to the lower level of the hierarchy (e.g., a disk).
27. **Q: Which write policy is typically used by lowest-level caches today, and why?**
A: Write-back, because the rate of processor writes generally exceeds the rate at which memory can process them.
28. **Q: What is the Three Cs model used for?**
A: It provides an intuitive model for classifying cache misses into three categories.
29. **Q: What are the three categories of misses in the Three Cs model?**
A: Compulsory misses, capacity misses, and conflict misses.
30. **Q: Define Compulsory Miss.**
A: A cache miss caused by the first access to a block that has never been in the cache; also called cold-start miss.
31. **Q: Define Capacity Miss.**
A: A cache miss that occurs because the cache cannot contain all the blocks needed during execution, even with full associativity.
32. **Q: Define Conflict Miss.**
A: A miss in a direct-mapped or set-associative cache when multiple blocks compete for the same set; would be eliminated in a fully associative cache.
33. **Q: What is the primary way to reduce compulsory misses?**
A: Increase the block size.
34. **Q: What is a straightforward way to reduce capacity misses?**
A: Enlarge the cache size.
35. **Q: How can conflict misses be reduced?**
A: By increasing the associativity.
36. **Q: What is the potential negative effect of increasing cache size?**
A: It may increase the cache access time.
37. **Q: What is the potential negative effect of increasing associativity?**
A: It may increase the cache access time.
38. **Q: What are the potential negative effects of increasing block size?**
A: It increases the miss penalty, and a very large block size could increase the overall miss rate.
39. **Q: Is it true that there is no way to reduce compulsory misses?**
A: No, this is false; increasing the block size can reduce compulsory misses.

40. **Q: Is it true that fully associative caches have no conflict misses?**
A: Yes, this is true by definition.
41. **Q: As cache sizes grow, the relative improvement from associativity -----, and the absolute improvement -----.**
A: Increases only slightly; shrinks significantly.
42. **Q: Why do smaller caches obtain larger absolute benefit from associativity?**
A: Because the base miss rate of a small cache is larger, providing more opportunity for improvement.
43. **Q: The method of finding a block is dictated by the ----- scheme.**
A: Block placement.
44. **Q: What enables on-chip L2 caches to have higher associativity?**
A: Their hit times are not as critical, and designers are not constrained by standard SRAM chips.
45. **Q: In a direct-mapped cache, how many candidates are there for replacement?**
A: Only one.
46. **Q: What hardware feature helps the OS approximate LRU in virtual memory?**
A: Reference bits or equivalent functionality to track page usage.
47. **Q: In the Three Cs model, the total miss rate of a direct-mapped cache includes conflict misses from -----.**
A: One-way, two-way, and four-way associativity components.

9.2 From Flashcards (21)

48. **Q: In a direct-mapped placement, how many sets and blocks per set are there?**
A: As many sets as blocks in the cache, with exactly one block per set.
49. **Q: In a fully associative placement, how many sets and blocks per set are there?**
A: There is only one set, which contains all the blocks in the cache.
50. **Q: As cache sizes grow, the ----- improvement from associativity shrinks significantly.**
A: Absolute.
51. **Q: How is a block found in a fully associative cache?**
A: By searching through all entries in the entire cache.
52. **Q: What mechanism is used to find a block (page) in virtual memory systems?**
A: A separate lookup table, known as the page table.
53. **Q: What is the first reason virtual memory uses fully associative placement?**
A: Full associativity is beneficial because memory misses are extremely expensive.
54. **Q: What is the second reason virtual memory uses fully associative placement?**
A: It allows software to use sophisticated replacement schemes to reduce the miss rate.
55. **Q: What is the third reason virtual memory uses fully associative placement?**
A: The full map (page table) can be easily indexed with no extra hardware or searching required.
56. **Q: For which types of caches must a replacement policy be decided?**
A: Set-associative and fully associative caches.
57. **Q: What are the two primary strategies for block replacement in associative caches?**
A: Random and Least Recently Used (LRU).
58. **Q: Define the LRU replacement policy.**
A: The block that is replaced is the one that has been unused for the longest time.
59. **Q: Why is some form of LRU always approximated for virtual memory?**
A: Because the cost of a miss is enormous, so even a tiny reduction in the miss rate is very important.
60. **Q: The ----- write policy involves writing to both the cache and lower level.**
A: Write-through.

61. **Q: The _____ write policy involves writing only to the cache, and to lower level only when replaced.**
A: Write-back.
62. **Q: What is an advantage of write-back concerning processor write speed?**
A: Individual words can be written at the speed of the cache, not the slower main memory.
63. **Q: What is an advantage of write-back concerning multiple writes to the same block?**
A: Multiple writes within a single block require only one write to lower-level memory when the block is finally replaced.
64. **Q: What is an advantage of write-through concerning cache misses?**
A: Misses are simpler and cheaper because they never require a modified block to be written back first.
65. **Q: Why is write-back the only practical policy for virtual memory?**
A: Because of the extremely long latency of a write to the lower level (e.g., a disk).
66. **Q: Which write policy do lowest-level caches typically use today and why?**
A: Write-back, because the rate of processor writes generally exceeds what memory can process.
67. **Q: A conflict miss would be eliminated if the cache were _____ of the same size.**
A: Fully associative.
68. **Q: What is the most direct way to reduce conflict misses?**
A: By increasing the associativity of the cache.
69. **Q: What is the most direct way to reduce capacity misses?**
A: By enlarging the total size of the cache.
70. **Q: What is a potential negative impact of increasing the block size?**
A: It increases the miss penalty (the time to fetch the larger block).
71. **Q: Do fully associative caches have conflict misses?**
A: No, by definition, fully associative caches have no conflict misses.
72. **Q: As associativity increases from 4-way to 8-way, what happens to improvement?**
A: There is very little improvement compared to the gain from 1-way to 2-way.
73. **Q: What is a key trade-off that makes memory hierarchy design challenging?**
A: Every change that can improve the miss rate (like increasing size or associativity) can also negatively affect performance (like increasing access time).
74. **Q: In the Three Cs graph, which component is so small it's often not visible?**
A: The compulsory miss component.
75. **Q: Why have first-level caches been growing slowly compared to second-level caches?**
A: To avoid increasing access time, which could lower overall performance, as L1 access time is critical to processor cycle time.
76. **Q: In a set-associative cache, how many blocks are candidates for replacement on a miss?**
A: All the blocks within the specific set where the miss occurred.
77. **Q: How is the location method for a block in a memory hierarchy determined?**
A: It is dictated by the block placement scheme (direct-mapped, set-associative, or fully associative).